(Please write your Exam Roll No.) Exam Roll No. .....

END TERM EXAMINATION

THIRD SEMESTER [B.TECH] JANUARY 2024

Paper Code: ES-201 **Subject: Computational Methods**

Time: 3 Hours **Maximum Marks: 75**

Note: Attempt five questions in all including Q.No.1 which is compulsory. Select one question from each unit. Assume missing data if any. Scientific calculator is allowed.

Q1 Attempt all questions:

a) The height of an observation tower was estimated to be 47m, whereas its actual height was 45m. Calculate the percentage relative error in the measurement. **(1.5)**

b) Derive the formula for evaluating $\sqrt{12}$ by Newton's Raphson method. **(1.5)**

c) Differentiate between partial and complete pivoting in solving linear system of algebraic equations. **(1.5)**

d) Estimate the missing term in the following table: **(1.5)**

x:	0	1	2	3	4
$f(x)$:	1	3	9	__	81

e) Define cubic spline function and state the conditions required for cubic spline interpolation. **(1.5)**

f) Find $\int_0^{\pi/2} \sqrt{\cos\theta}\, d\theta$ using Simpson's 1/3rd rule for $n=6$. **(2.5)**

g) Solve the IVP $\frac{dy}{dx} = xe^y, y(0)=0$ using Picard's method and estimate $y(0.2)$. **(3)**

h) Determine which of the following equations are elliptic, parabolic, and hyperbolic. **(2)**

(i) $f_{xx}+6f_{xy}+9f_{yy}=0$

(ii) $f_{xx}-2f_{xy}+2f_{yy}=2x+5y$

UNIT-I

Q2 a) Find the root of the equation $x=e^{-x}$, correct to three decimal places by Secant method by performing six iterations. **(8)**

b) Use Newton—Rapshon method to obtain a root of $\sin x = 1-x$ to three decimal places. **(7)**

Q3 a) Perform four iterations of Golden section search method to minimize $f(x)=x^4-14x^3+60x^2-70x$, $x\in(0,2)$ with $\varepsilon=10^{-3}$. **(7.5)**

b) Use steepest descent method for 3 iterations on $f(x_1,x_2,x_3)=(x_1-4)^4+(x_2-3)^2+4(x_3+5)^4$ with initial point $x^{(0)}=[4,2,-1]^T$ **(7.5)**

UNIT-II

Q4 a) Prove that $\Delta=\frac{1}{2}\delta^2+\delta\sqrt{(1+\delta^2/4)}$. **(7)**

b) The table gives the distance in nautical miles of the visible horizon for the given heights in feet above the earth's surface: **(8)**

P.T.O.

(Please write your Enrolment No. immediately) Enrolment No. _____

MID TERM EXAMINATION

B.TECH PROGRAMMES (UNDER THE AEGIS OF USICT)

3rd Semester, November, 2022

Paper Code: ES-201 **Subject: Computational Methods**
Time: 1½Hrs. **Max. Marks: 30**

Note: Attempt Q.No.1 which is compulsory and any two more questions from remaining.

Q.No.	Questions	Max. Marks	CO(s)
1(a)	Find a root of the equation $x^3 - 4x - 9 = 0$ using Bisection method.	2.5	CO1
1(b)	Define the Absolute, Relative and Truncation Errors.	2.5	CO1
1(c)	Evaluate $\Delta^3[(1-6x)(6-x)(5-3x)]$	2.5	CO2
1(d)	Use Trapezoidal rule to estimate the integral $\int_0^1 \frac{dx}{1+x}$.	2.5	CO2
2(a)	Perform four iteration of the Newton's Raphson method to obtain the approximate value of $(17)^{1/3}$ starting with the initial approximation $x_0=2$.	5.0	CO1
2(b)	Minimize $f(x) = x^2 + \frac{54}{x}$ in interval (0,5). Using Fibonacci search method.	5.0	CO1
3(a)	For the data X :- -2 -1 0 1 2 3 F(x):- 15 5 1 3 11 25 Construct the Newton's forward difference table and find f(-1.5).	5.0	CO2
3(b)	Find the value of f(9) from the given table X: 5 7 11 13 17 F(X) 150 392 1492 2366 5202	5.0	CO2
4(a)	Round off the numbers 86.5250 and 37.46235 to four significant figures and compute Absolute error, Relative Error and Percentage error in each case.	5.0	CO1
4(b)	Find the Polynomial of possible degree which assume the value 3,12,15,-21 when x has the value 3,2,1,-1 respectively.	5.0	CO2